

# Understanding and Improving Source-free Domain Adaptation from a Theoretical Perspective

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## Abstract

Source-free Domain Adaptation (SFDA) is an emerging and challenging research area that addresses the problem of unsupervised domain adaptation (UDA) without source data. Though numerous successful methods have been proposed for SFDA, a theoretical understanding of why these methods work well is still absent. In this paper, we shed light on the theoretical perspective of existing SFDA methods. Specifically, we find that SFDA loss functions comprising discriminability and diversity losses work in the same way as the training objective in the theory of self-training based on the expansion assumption, which shows the existence of the target error bound. This finding brings two novel insights that enable us to build an improved SFDA method comprising 1) Model Training with Auto-Adjusting Diversity Constraint and 2) Augmentation Training with Teacher-Student Framework, yielding a better recognition performance. Extensive experiments on three benchmark datasets demonstrate the validity of the theoretical analysis and our method.

## 1. Introduction

Deep learning has suffered from the domain-shift problem where models perform well on domains seen in the training phase but struggle with unseen domains. Unsupervised domain adaptation (UDA) is a promising solution: it transfers knowledge learned from a labeled source domain to an unlabeled target domain. UDA methods show their effectiveness on various computer vision tasks such as classification [25, 26, 54], object detection [6, 37, 59], segmentation [15, 16, 50], etc.; however, they typically require both source and target domain data, which limits their applicability as this requirement poses privacy concerns about the source data and entails computational inefficiency. Recently, researchers have shifted focus to another direction of UDA called source-free domain adaptation (SFDA). SFDA bypasses the above issues by not using raw data from the

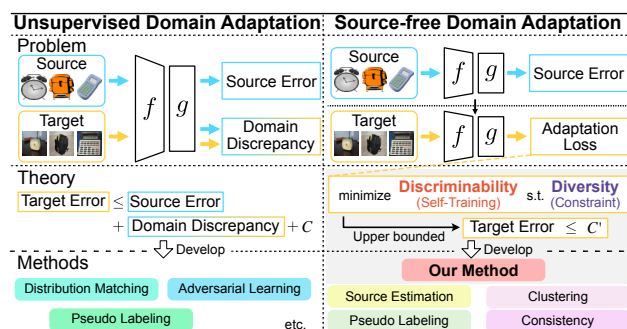


Figure 1. **Overview of our work.** Unsupervised domain adaptation has a theoretical background that has yielded a variety of methods. By contrast, the theoretical perspectives on source-free domain adaptation (SFDA) have not been well explored. Our research motivations are (1) to shed light on the theoretical perspectives of existing SFDA methods, and (2) to propose an improved method based on the theoretical insights.

source data. Instead, SFDA performs the training with a source-pre-trained model and unlabeled target data. Various SFDA methods have been proposed, including source estimation [11, 17, 33, 42, 49], pseudo-labeling [19, 34, 51], clustering [21–23], consistency [5, 52, 53, 57], and even without source data, they outperform UDA methods.

Despite these promising achievements, a theoretical understanding of SFDA methods is still lacking. As shown in Fig. 1, UDA studies rely on the theoretical notion that the target error can be upper-bounded by the source error and the discrepancy between the two domains [3, 30, 55, 58], and develop various approaches such as distribution matching [26, 27, 44], adversarial learning [12, 28, 45, 54], and pseudo-labeling [20, 25, 60, 61]. By contrast, the theoretical analyses of SFDA are either absent or not general enough and cannot form the basis for the development of new methods. Moreover, the theory of UDA is not directly applicable to SFDA due to inaccessibility of source data.

In this paper, we shed light on the theoretical perspective of existing SFDA methods through the theory of self-training based on the expansion assumption [48]. Self-training is an approach that utilizes the current model per-

dictionaries of the unlabeled data for further training, and the expansion assumption states that the data distribution has good continuity within each class. The theory asserts that, under the expansion assumption, there is an upper bound on the target error when the model is trained on the objective with a **self-training** term encouraging prediction consistency among the augmented unlabeled samples and a **constraint** term ensuring prediction diversity. We reveal an interesting correspondence between this training objective and the SFDA training loss. Recent studies [9, 53] have discovered a feature common that most SFDA methods employ the combination of **discriminability** and **diversity** losses: the former improves the model discriminability to the unlabeled target samples while the latter ensures predictions for all classes. As illustrated in the middle right of Fig. 1, we find that the discriminability and diversity losses perform the same respective roles as the self-training term and the constraint term of the theory, which provides us the theoretical understanding of SFDA. In addition, our analysis brings the following new insights: 1) the trade-off between discriminability and diversity should be adjusted as training progresses, and 2) the upper bound of the target error depends on how we design the data augmentation.

Based on the above insights, we propose an improved SFDA method incorporating 1) Model Training with Auto-Adjusting Diversity Constraint and 2) Augmentation Training with Teacher-Student Framework. In the former training, we update the model on the basis of the discriminability and diversity losses while introducing a novel technique to automatically adjust the trade-off parameter between discriminability and diversity. In the latter training, we introduce a learnable data augmentation and update its parameters by using the predictions of the current model and the teacher model, yielding a tighter upper bound. Experimental results with three benchmarks (Office-31 [36], Office-Home [47], VisDA2017 [32]) show the validity of our theoretical analysis and the proposed method.

In summary, our contributions are: i) by using the theory of self-training based on the expansion assumption [48], we reveal that a model trained with discriminability and diversity losses will achieve a small target error; ii) we propose an improved SFDA method incorporating Model Training with Auto-Adjusting Diversity Constraint and Augmentation Training with Teacher-Student Framework.

## 2. Related Works

**Unsupervised Domain Adaptation (UDA).** On the basis of the theoretical foundation that the target error is upper bounded by the source error and the distributional discrepancy between the two domains [2, 3, 30, 55, 58], various UDA methods have been developed. Distribution matching approaches [26, 27, 44] directly minimize the measures of distribution discrepancy (e.g. maximum mean discrepancy

(MMD)). Adversarial learning approaches [12, 28, 45, 54] reduce the discrepancy by learning domain-invariant representations using an additional domain classifier. Pseudo-labeling approaches [20, 25, 60, 61] not only minimize the domain discrepancy but also improve the feature discriminability by using the pseudo-labeled target samples.

Although the theory of UDA has yielded various methods, it is not applicable to SFDA due to the inaccessibility of the source data. In this study, we instead employed the theory of self-training based on the expansion assumption [48] as a way to understand SFDA methods.

**Source-free Domain Adaptation (SFDA).** With reference to [24], SFDA methods can be roughly categorized into four approaches. Source-estimation approaches [11, 17, 22, 33, 42, 49] generate pseudo-source data using a pre-trained model, which transforms the SFDA problem into a conventional UDA problem. Pseudo-labeling approaches [19, 34, 51] assign a class label to each unlabeled target sample using the current model and use them in a supervised manner. Based on the cluster assumption [46], clustering approaches [21–23] encourage minimizing the uncertainty of the model predictions or performing clustering over the target features. Inspired by the consistency regularization of semi-supervised learning [4, 39, 41], consistency approaches [5, 52, 53, 57] train the model to maximize the prediction consistency regardless of the perturbations on the input data or the model parameters.

Despite many successful SFDA methods, most do not have a theoretical foundation yet, except for the source estimation approaches that convert SFDA to conventional UDA. A few studies [57] have theoretically investigated their methods, but they are not applicable to the others. In this paper, we introduce the theory of self-training based on the expansion assumption [48] to give a theoretical perspective on a wide range of SFDA methods, and we propose an improved method based on our theoretical analysis.

## 3. Understanding SFDA from a Theoretical Perspective

In this section, we define the SFDA problem, and then, see the common feature of SFDA methods. Finally, we provide a theoretical analysis of SFDA through [48].

### 3.1. Problem Definition

The upper right of Fig. 1 illustrates the SFDA problem definition. We are given a model  $F_S : \mathcal{X} \rightarrow \mathcal{Y}$  trained on a labeled source domain data  $D_S$  and an unlabeled target domain data  $D_T = \{x_t^{(i)}\}_{i=1}^{n_t}$ . Generally, the model  $F$  consists of a feature extractor  $f$  and a fully-connected layer based classifier  $g$ . The goal is to train the model  $F$  to obtain a target-adapted model  $\hat{F}$  that has low target error without using source domain data  $D_S$  nor target labels  $y_t^{(i)}$ .

### 3.2. Common Feature of SFDA Methods

Whilst lacking a precise theoretical background, the following observation [9, 53] provides a key to understanding SFDA methods: the existing SFDA methods have a common feature that their training loss functions can be decomposed into **discriminability** and **diversity** losses wherein the discriminability loss enhances the model discriminability to the unlabeled target samples while the diversity loss ensures the model has predictions for diverse classes.

For example, SHOT-IM [23], a pioneering work on SFDA, trains the model on the basis of mutual information maximization, i.e., a training loss function comprising conditional entropy minimization (discriminability) and marginal entropy maximization (diversity):

$$\mathcal{L}_{\text{MIM}} = \underbrace{H(Y|X)}_{\text{discriminability}} - \lambda_{\text{div}} \underbrace{H(Y)}_{\text{diversity}}. \quad (1)$$

where  $\lambda_{\text{div}}$  represents the trade-off parameter of the loss.

Another example is AaD [53], which trains the model by maximizing the prediction similarities among the local neighborhoods in the feature space (discriminability) while minimizing those of the others (diversity) as follows:

$$\mathcal{L}_{\text{AaD}} = \frac{1}{|\mathbf{B}|} \sum_{i \in \mathbf{B}} \left\{ \underbrace{\frac{1}{|\mathbf{C}_i|} \sum_{\hat{\mathbf{p}} \in \mathbf{C}_i} -\mathbf{p}_i \cdot \hat{\mathbf{p}}}_{\text{discriminability}} + \lambda_{\text{div}} \underbrace{\sum_{j \in \mathbf{B} \setminus \{i\}} \mathbf{p}_i \cdot \mathbf{p}_j}_{\text{diversity}} \right\}, \quad (2)$$

where  $\mathbf{B}$  is a mini-batch,  $\mathbf{p}_i$  is the prediction of sample  $i$ , and  $\mathbf{C}_i$  is a set of the predictions of  $K$ -nearest neighborhoods of sample  $i$  on the feature space.

This common feature is widely seen in other methods, such as those using pseudo-labeling or consistency regularization as their own discriminability loss with the above marginal entropy maximization (diversity) [34, 51, 52], and those employing contrastive learning, which maximizes the similarity of positive pairs (discriminability) and minimizes the similarity of negative pairs (diversity) [5, 19, 57].

Now that we have confirmed the discriminability and diversity losses to be the key to the success of SFDA methods, but *why are these losses crucial for the success of SFDA?*

### 3.3. Theoretical Understanding of SFDA

We will answer the above question by introducing the theory of self-training based on the expansion assumption [48]. This theory shows that, under certain assumptions, the model will have a low target error when it is **self-trained** based on prediction consistency while a **constraint** is imposed to ensure prediction diversity.

**Notations.** We let  $\mathcal{A}$  denote the family of data augmentation and define an augmented sample set an input  $x$  as  $\mathcal{B}(x) := \{x' \mid \exists A \in \mathcal{A} \text{ s.t. } \|x' - A(x)\| < r\}$ , the neighborhoods of  $x$  as  $\mathcal{N}(x) := \{x' \mid \mathcal{B}(x) \cap \mathcal{B}(x') \neq \emptyset\}$ ,

the neighborhoods of the set  $V$  as  $\mathcal{N}(V) := \cup_{x \in V} \mathcal{N}(x)$ , and prediction inconsistency for a  $C$ -class prediction model  $F : \mathcal{X} \rightarrow [C]$  on a distribution  $P$  as  $R_{\mathcal{B}}(F) := \mathbb{E}_P[\mathbf{1}(\exists x' \in \mathcal{B}(x) \text{ s.t. } F(x') \neq F(x))]$ .

**Assumptions.** Before explaining the main theorem that is the key to our theoretical analysis, we must make two assumptions: *Expansion* and *Separation*<sup>1</sup>.

*Expansion* assumes that the data of the same class are distributed in a continuous region and that any small region  $V$  has a neighborhood region of the same class larger than  $V$ . Concretely,  $P_i(\mathcal{N}(V)) \geq dP_i(V)$  for  $P_i(V) \leq 1/2$ ,  $d > 1$ , where  $P_i(V)$  represents the proportion of subset  $V$  in the total class  $i$  data and  $d$  represents an expansion factor that corresponds to the strength of the data augmentation.

*Separation* assumes that the distribution of different classes is separated and the predictions of the ground-truth model  $F^*$  will not be altered by the data augmentation, i.e.,  $R_{\mathcal{B}}(F^*) < \mu$ , where  $\mu$  represents a negligible value.

**Theory for Understanding SFDA.** With the above assumptions, [48] derives the following theorem that is key to understanding why existing SFDA methods perform well.

**Theorem 1.** Suppose that the above two assumptions hold for some  $d, \mu$  such that  $\min_{y \in [C]} P(\{x : F^*(x) = y\}) > \max\{2/(d-1), 2\}\mu$ . Then any minimizer  $\hat{F}$  of

$$\underbrace{\min_F R_{\mathcal{B}}(F)}_{\text{self-training} = \text{discriminability}} \quad \text{subject to} \quad \underbrace{\min_{y \in [C]} \mathbb{E}_P[\mathbf{1}(F(x) = y)] > \max\left\{\frac{2}{d-1}, 2\right\} R_{\mathcal{B}}(F)}_{\text{constraint} = \text{diversity}} \quad (3)$$

satisfies

$$\text{Err}_{\mathcal{U}}(\hat{F}) \leq \max\left\{\frac{d}{d-1}, 2\right\}\mu, \quad (4)$$

where  $\text{Err}_{\mathcal{U}}(\hat{F}) := \min_{\pi: [C] \rightarrow [C]} [\mathbf{1}(\pi(F(x)) \neq F^*(x))]$ , and  $\pi$  represents a permutation.

Theorem 1 shows that a model trained with the objective function (3) consisting of a self-training term that requires neighborhood predictions to be consistent and a constraint term that assigns a certain portion of predictions to all classes has an upper bound on the target error (4).

**What we want to highlight is that the SFDA training loss with discriminability and diversity loss acts the same as the objective (3).** Specifically, increasing the **discriminability** leads to a reduction in the prediction inconsistency of the **self-training** term while the **diversity** loss functions in the same way as the **constraint** term that ensures the minor class predictions. This indicates that we can apply Theorem 1 to the SFDA training loss, and thus, the SFDA-trained model also has an upper bound on the target error.

<sup>1</sup>Formal statements of the assumptions are given in Appendix A.

| Method                 | Acc [%]     |
|------------------------|-------------|
| i) Source only         | 72.4        |
| ii) Dis only           | 93.0        |
| iii) Dis + Div         | 95.3        |
| iv) Dis + Div w/ Decay | <b>95.7</b> |

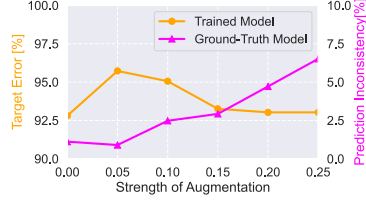


Table 1. Model accuracy w/ and w/o discriminability and diversity losses.

Figure 2. Model accuracy ( $\text{Err}_U(\hat{F})$ ) and the prediction inconsistency of the ground-truth model ( $\mu$ ) against strength of augmentation ( $d$ ).

**Preliminary Experiment with Synthetic Data.** We verified this correspondence through an experiment on a two-dimensional synthetic dataset. We employed a variant of the inter-twinning moons 2D dataset, where we simulated the domain shift by rotation. We used the training objective of SHOT-IM [23] and 2D-Gaussian perturbations as the data augmentation. In the experiment, we compared the accuracy of i) Source only, ii) Discriminability (Dis) only, and iii) Discriminability and Diversity (Dis + Div). Moreover, we measured the accuracy of the trained model  $\text{Err}_U(\hat{F})$  and the prediction inconsistency of the ground-truth model corresponding to  $\mu$  versus the strength of the augmentation corresponding to  $d$ . Other details on the experimental settings are described in Appendix B.

The results are summarized in Tab. 1 and Fig. 2. Tab. 1 shows that the accuracy is improved by incorporating the diversity loss in addition to the discriminability loss, which establishes the necessity of the diversity loss to function as the constraint of the objective (3). The prediction accuracy of the trained model in Fig 1 aligns with the upper bound (4); namely, as shown in Fig. 2, by keeping the prediction inconsistency of the ground-truth model  $\mu$  low and increasing the strength of the data augmentation  $d$ , the model can achieve a low target error.

**Theoretical Insights.** Besides, Theorem 1 provides two insights for the further improvement of SFDA methods.

First, the weight of the diversity loss  $\lambda_{\text{div}}$  should be adjusted as training progresses, and this can be done by controlling the value of the discriminability loss. The right-hand side (RHS) of the constraint in the objective (3) includes  $R_B(F)$ , which decreases during the training. This indicates that it is more reasonable to let the constraint decay along with  $R_B(F)$ , i.e., the discriminability loss. The results shown in Tab 1 also demonstrate that decaying  $\lambda_{\text{div}}$  (Dis + Div w/ Decay) brings a better result. However, most of the existing SFDA methods fix  $\lambda_{\text{div}}$ , which would be sub-optimal. Although AaD [53] exceptionally uses a manually designed scheduler, tuning it is laborious.

Second, the upper bound of the target error depends on the parameters  $d$  and  $\mu$  which are relevant to the data augmentation properties. This indicates that how we design the

data augmentation is a critical factor in training models with better accuracy. However, the prior studies on SFDA [5, 57] have paid less attention to it and have used pre-defined data augmentations [7, 8], which may not be optimal for SFDA.

## 4. Our Method: Improved SFDA based on Theoretical Insights

Fig. 3 shows the overview of our method, which has three major components: a prediction model  $F = g \circ f$ , a teacher model  $F' = g' \circ f'$ , and a learnable augmentation  $\mathcal{A}$ . On the basis of the above insights, we developed an improved SFDA method comprising 1) Model Training with Auto-Adjusting Diversity Constraint, and 2) Augmentation Training with Teacher-Student Framework.

### 4.1. Model Training with Auto-Adjusting Diversity Constraint

We update  $F$  upon the modified discriminability and diversity losses of AaD [53], coupled with a novel technique to automatically adjust the trade-off parameter between discriminability and diversity.

**Discriminability and Diversity Losses.** Considering that the prediction inconsistency  $R_B(F)$  in the self-training term is originally calculated among data-augmented samples, we modify the training loss (2) so as to calculate the prediction dissimilarity among data-augmented samples. The discriminability and diversity losses are formally defined as

$$\mathcal{L}_{\text{dis}} = \frac{1}{M|\mathbf{B}||\mathbf{C}_i|} \sum_{i \in \mathbf{B}} \sum_{\mathbf{p} \in \mathbf{C}_i} \sum_{m=1}^M (1 - \mathbf{p}_i^m \cdot \hat{\mathbf{p}}), \quad (5)$$

$$\mathcal{L}_{\text{div}} = \frac{1}{M|\mathbf{B}|} \sum_{i \in \mathbf{B}} \sum_{j \in \mathbf{B} \setminus \{i\}} \sum_{m=1}^M \mathbf{p}_i^m \cdot \mathbf{p}_j, \quad (6)$$

where  $\mathbf{p}_i^m = F(A_m(x_t^{(i)}))$ ,  $A_m$  is the  $m$ -th augmentation sampled from  $\mathcal{A}$ ,  $M$  is the number of augmentations. To retrieve the  $K$ -nearest neighbors  $\mathbf{C}_i$  efficiently, we build a memory bank that stores the feature vectors  $\mathbf{z}_i = f(x_t^{(i)})$  and predictions  $\mathbf{p}_i = F(x_t^{(i)})$  of all target samples in  $D_T$ .

**Auto-Adjusting Diversity Constraint.** As shown in Sec. 3.3,  $\lambda_{\text{div}}$  should be adjusted as the training progresses. Specifically, the RHS of the constraint of the objective (3) involves the prediction inconsistency  $R_B(F)$  which will get smaller as the training progresses. Moreover, the discriminability loss  $\mathcal{L}_{\text{dis}}$  functions the same way as  $R_B(F)$ . This means that we can easily control  $\lambda_{\text{div}}$  with  $\mathcal{L}_{\text{dis}}$ . Accordingly, the auto-adjusting  $\lambda_{\text{div}}$  can be simply expressed as

$$\lambda_{\text{div}} = \lambda_{\text{div}}^{\max} \mathcal{L}_{\text{dis}}, \quad (7)$$

where  $\lambda_{\text{div}}^{\max}$  determines the maximum size of  $\lambda_{\text{div}}$ . Note that we apply the stop-gradient operation to  $\lambda_{\text{div}}$ .



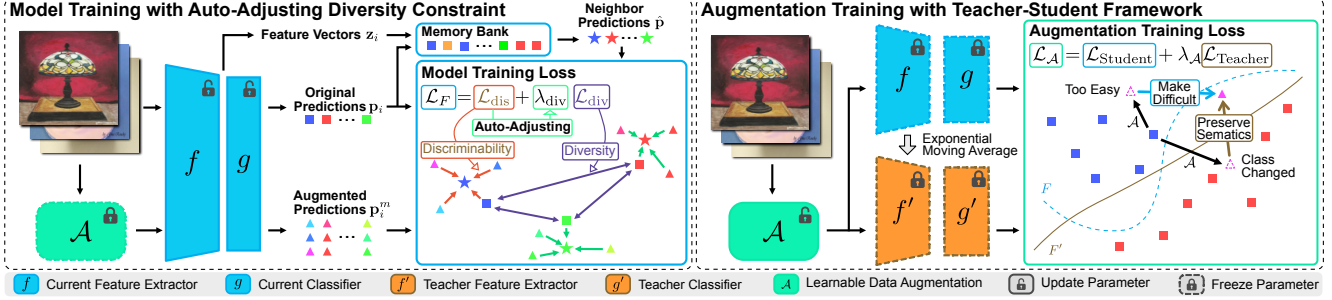


Figure 3. **Overview of our method.** In Model Training with Auto-Adjusting Diversity Constraint, we train the model by minimizing the discriminability loss  $\mathcal{L}_{\text{dis}}$  and the diversity loss  $\mathcal{L}_{\text{div}}$  while automatically adjusting the trade-off parameter between discriminability and diversity. In Augmentation Training with Teacher-Student Framework, we train a learnable data augmentation  $\mathcal{A}$  to generate harder samples for the current model  $F$  while suppressing the prediction inconsistency of the teacher model  $F'$ .

**Model Training.** The training loss of the model using the learnable augmentation  $\mathcal{A}$  is

$$\mathcal{L}_F = \mathcal{L}_{\text{dis}} + \lambda_{\text{div}} \mathcal{L}_{\text{div}}. \quad (8)$$

However, we find in an early study that the learnable data augmentation  $\mathcal{A}$  (whose details will be described in Sec. 4.2) yields some heavy augmentations (e.g., rotation, invert, etc.), which may impair the model training. Inspired by [1], we stabilize the training by incorporating a loss  $\mathcal{L}'_F$  calculated among samples with weak augmentations (e.g., random\_clip, random\_flip).

In summary, the total loss of the model training is

$$\min_F \mathcal{L}_F^{\text{Total}} = \lambda_F \mathcal{L}_F + (1 - \lambda_F) \mathcal{L}'_F. \quad (9)$$

where  $\lambda_F$  is a hyper-parameter to control the loss balance.

## 4.2. Augmentation Training with Teacher-Student Framework

As discussed in Sec. 3.3, the upper bound of the target error depends on how the data augmentation is designed. Motivated by [40], we update the learnable data augmentation  $\mathcal{A}$  in the teacher-student framework to get a tighter bound.

**Learnable Data Augmentation.**  $\mathcal{A}$  consists of  $L$  different augmentations  $A^{(l)}$  ( $l = 1, 2, \dots, L$ ). A single augmentation consists of  $N$  consecutive transformation operations  $O_1^{(l)}, \dots, O_N^{(l)}$ . Each operation includes affine transformations (e.g. shear\_x) and color enhancing operations (e.g. contrast), and it has a magnitude parameter  $m_n^{(l)} \in [0, 1]$  to control the transformation strength and a probability parameter  $p_n^{(l)} \in [0, 1]$  to control whether to apply the operation. To facilitate parameter optimization, we utilize Faster AutoAugment [13] to make these parameters differentiable and updatable by gradient descent.

**Augmentation Training.** We train  $\mathcal{A}$  to make the bound tighter based on the observation that the upper bound of the target error (4) becomes tighter as 1) the strength of the augmentation  $d$  gets larger and 2) the prediction inconsistency

of the ground-truth model  $\mu$  gets smaller. Since we cannot actually access the ground-truth model  $F^*$ , we use the teacher-student framework and assign the teacher model to be a proxy for  $F^*$ ; 1) we increase  $d$  by training  $\mathcal{A}$  to augment samples that are harder for the current (student) model to predict, and 2) we decrease  $\mu$  by training  $\mathcal{A}$  to augment samples that are recognizable to the teacher model.

More specifically, 1) we train  $\mathcal{A}$  to maximize the prediction entropy of the current model:

$$\mathcal{L}_{\text{Student}} = -\frac{1}{|\mathbf{B}|} \sum_{i \in \mathbf{B}} \sum_{c=1}^C p_i[c] \log(1 - p_i[c]), \quad (10)$$

where  $p_i[c] = F(A(x_t^{(i)}))[c]$  is the prediction probability of sample  $i$  for class  $c$ , and  $A$  is a augmentation operation randomly sampled from  $\mathcal{A}$ . Following [40], we employ a non-saturating prediction entropy instead of the naive one.

Whereas, 1) we train  $\mathcal{A}$  to minimize the prediction entropy of the teacher model:

$$\mathcal{L}_{\text{Teacher}} = -\frac{1}{|\mathbf{B}|} \sum_{i \in \mathbf{B}} \sum_{c=1}^C p'_i[c] \log p'_i[c], \quad (11)$$

where  $p'_i[c] = F'(A(x_t^{(i)}))[c]$ . The teacher model is updated using the exponential moving average strategy [41]:

$$F' \leftarrow (1 - \beta)F + \beta F', \quad (12)$$

where  $\beta$  is a momentum parameter.

The total loss of the augmentation training is

$$\min_{\mathcal{A}} \mathcal{L}_{\mathcal{A}} = \mathcal{L}_{\text{Student}} + \lambda_{\mathcal{A}} \mathcal{L}_{\text{Teacher}}, \quad (13)$$

where  $\lambda_{\mathcal{A}}$  is a coefficient hyper-parameter.

**Discussion.** Our method is different from the teacher-student-based SFDA methods [5, 56] in that we use the teacher-student framework for training data augmentation. Our approach may appear similar to the prior work [43] but fundamentally differs in that ours treats the teacher model as a proxy of the ground truth model based on our theoretical insights.

**Algorithm 1** Training Procedure of Our Method.**Input:** Source-trained model  $F_s$ , Target domain data  $D_T$ **Output:** Target-adapted model  $\hat{F}$ 

```

1: for  $e$  in  $\{1 \cdots E\}$  do
2:   for Mini-batch  $\mathbf{B}$  in  $D_T$  do ▷ Model Training
3:     Calculate loss  $\mathcal{L}_F^{\text{Total}}$  on  $\mathbf{B}$ 
4:     Update  $F$  to minimize  $\mathcal{L}_F^{\text{Total}}$ 
5:     Update  $F'$  based on (12)
6:   end for
7:   if  $e \equiv 0 \pmod{\hat{e}}$  then
8:     for Mini-batch  $\mathbf{B}$  in  $D_T$  do ▷ Augmentation Training
9:       Calculate loss  $\mathcal{L}_A$  on  $\mathbf{B}$ 
10:      Update  $\mathcal{A}$  to minimize  $\mathcal{L}_A$ 
11:    end for
12:  end if
13: end for

```

### 4.3. Training Procedure of Our Method

As shown Algorithm 1, our method alternately performs model training and data augmentation training. To control the speed of these pieces of training, we set a parameter  $\hat{e}$  that determines the interval of the augmentation training.

## 5. Experiments

We experimentally compared the performance of our method and the existing SFDA methods on three benchmark datasets, and we examined the validity of our theoretical insights through further analyses.

### 5.1. Setups

**Datasets.** We used three benchmark datasets: *Office-31* [36] is a small-scale dataset, which consists of 31 categories and 3 domains (Amazon, Webcam and Dslr). *Office-Home* [47] is a moderate-scale dataset, which consists of 65 categories and 4 domains (Art, Clipart, Product and Real world). *VisDA2017* [32] is a large synthetic-to-real adaptation benchmark dataset with 12 categories. For *Office-31* and *Office-Home*, we evaluated the accuracy on all source-target combinations, while we computed the average of per-class accuracies for *VisDA2017*.

**Network Architectures & Augmentation Implementations.** We use ResNet-50 [14] as the backbone network in the *Office-31* and *Office-Home* experiments, and ResNet-101 [14] in *VisDA2017*. All of the networks are pre-trained on Imagenet [10]. Following [23], we replaced the output layer of the backbone network with the following networks: a *fully-connected layer*  $\rightarrow$  *batch normalization* [18]  $\rightarrow$  *fully-connected layer with weight normalization* [38]. We implemented the learnable data augmentation  $\mathcal{A}$  with a public library of differentiable data augmentation<sup>2</sup>. We set  $L$  to 25, and  $N$  to 2 in all of the experiments.

<sup>2</sup><https://github.com/moskomule/dda>

Table 2. **Classification Accuracy (%) on Office-31 (ResNet-50).** The best and second best are highlighted in **bold** and with underline.

| Method (Source $\rightarrow$ Target) | A $\rightarrow$ D | A $\rightarrow$ W | D $\rightarrow$ W | W $\rightarrow$ D | D $\rightarrow$ A | W $\rightarrow$ A | Avg.               |
|--------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|--------------------|
| 3C-GAN [22]                          | 92.7              | 93.7              | 98.5              | 99.8              | 75.3              | 77.8              | 89.6               |
| SHOT [23]                            | 94.0              | 90.1              | 98.4              | <u>99.9</u>       | 74.7              | 74.3              | 88.6               |
| VDM-DA [42]                          | 94.1              | 93.2              | 98.0              | <b>100.0</b>      | 75.8              | 77.1              | 89.7               |
| A <sup>2</sup> Net [49]              | 94.5              | 94.0              | <b>99.2</b>       | <b>100.0</b>      | <u>76.7</u>       | 76.1              | 90.1               |
| NRC [51]                             | 96.0              | 90.8              | 99.0              | <b>100.0</b>      | 75.3              | 75.0              | 89.4               |
| CPGA [33]                            | 94.4              | 94.1              | 98.4              | 99.8              | 76.0              | 76.6              | 89.9               |
| CoWA-JMDS [21]                       | 94.4              | <b>95.2</b>       | 98.5              | 99.8              | 76.2              | <u>77.6</u>       | <u>90.3</u>        |
| C-SFDA [19]                          | <u>96.2</u>       | 93.9              | 98.8              | 99.7              | <b>77.3</b>       | <b>77.9</b>       | <b>90.5</b>        |
| AaD [53]                             | <b>96.4</b>       | 92.1              | <u>99.1</u>       | <b>100.0</b>      | 75.0              | 76.5              | 89.9               |
| <b>Improved SFDA</b>                 | 95.3              | <u>94.2</u>       | 98.3              | <u>99.9</u>       | 76.4              | 77.5              | <u>90.3 (+0.4)</u> |

**Source Training.** We use Nesterov SGD with a mini-batch size of 64 as the optimization algorithm on all three datasets.

For *Office-31* and *Office-Home*, we set the learning rate  $\eta$  to  $1e-2$  for the last replaced layers and  $1e-3$  for the backbone layers, momentum to 0.9, and weight decay to  $5e-4$ . We used a standard cross entropy loss with label smoothing for training. The label smoothing parameter was set to 0.1. We trained the model for 50 epochs.

For *VisDA2017*, we set the initial learning rate  $\eta_{\text{init}}$  to  $1e-3$  for the last replaced layers and  $1e-4$  for the backbone layers, momentum to 0.9, and weight decay to  $1e-3$ . The learning rate  $\eta$  was scheduled as;  $\eta = \eta_{\text{init}}(1 + 10p)^{-0.75}$ , where  $p$  was linearly increased from 0.0 to 1.0 throughout the training. We used the training losses of *Office-31* and *Office-Home*. We trained the model for 10 epochs.

**Target Training.** We use the Nesterov SGD for training  $F$  and AdamW [29] for training  $\mathcal{A}$  with mini-batch size 64.

For *Office-31* and *Office-Home*, we set Nesterov SGD parameters as follows: the learning rate  $\eta$  for the second last layer to  $3e-3$  and for the backbone layers to  $3e-4$ , momentum to 0.9, and weight decay to  $5e-4$ . We fixed the last layer during the training, which yielded better results. We set the parameters of AdamW to the Pytorch default values [31] except for the learning rate  $\eta^{\text{aug}}$  to  $5e-4$ . The number of the training epoch  $E$  was set to 100, and the interval  $\hat{e}$  was set to three. The other parameters were set as follows;  $\lambda_{\text{div}}^{\text{max}}$  to 0.4 for *Office-31* and 0.75 for *Office-Home*,  $\lambda_F$  to 0.5,  $\lambda_A$  to 1.0,  $K$  to two,  $M$  to three, and  $\beta$  to 0.99. We initialized  $\mathcal{A}$  with AutoAugment [7] Imagenet Policies.

For *VisDA2017*, we set the Nesterov SGD parameters as follows: the initial learning rate  $\eta_{\text{init}}$  for the last two layers to  $2.5e-3$  and for the backbone layers to  $2.5e-4$ , momentum to 0.9, and weight decay to  $1e-4$ . We set the initial learning rate of AdamW  $\eta_{\text{init}}^{\text{aug}}$  to  $2e-3$  and the other parameters are set as default.  $\eta$  and  $\eta^{\text{aug}}$  are scheduled as in the source training. The training epoch  $E$  was set to 100 and the interval  $\hat{e}$  to three. The other parameters were set as follows;  $\lambda_{\text{div}}^{\text{max}}$  to 0.08,  $\lambda_F$  to 0.2,  $\lambda_A$  to 1.0,  $K$  to five,  $M$  to three, and  $\beta$  to 0.99. We randomly initialized  $\mathcal{A}$ .

Table 3. **Classification Accuracy (%) on Office-Home (ResNet-50)**. The best and second best are highlighted in **bold** and with underline.

| Method (Source → Target) | Ar→Cl       | Ar→Pr       | Ar→Rw       | Cl→Ar       | Cl→Pr       | Cl→Rw       | Pr→Ar       | Pr→Cl       | Pr→Rw       | Rw→Ar       | Rw→Cl       | Rw→Pr       | Avg.               |
|--------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------------|
| SHOT [23]                | 57.1        | 78.1        | 81.5        | 68.0        | 78.2        | 78.1        | 67.4        | 54.9        | 82.2        | 73.3        | 58.8        | 84.3        | 71.8               |
| A <sup>2</sup> Net [49]  | 58.4        | 79.0        | 82.4        | 67.5        | 79.3        | 78.9        | <b>68.0</b> | 56.2        | 82.9        | <u>74.1</u> | 60.5        | 85.0        | 72.8               |
| G-SFDA [52]              | 57.9        | 78.6        | 81.0        | 66.7        | 77.2        | 77.2        | 65.6        | 56.0        | 82.2        | 72.0        | 57.8        | 83.4        | 71.3               |
| NRC [51]                 | 57.7        | <b>80.3</b> | 82.0        | 68.1        | 79.8        | 78.6        | 65.3        | 56.4        | 83.0        | 71.0        | 58.6        | 85.6        | 72.2               |
| CPGA [33]                | 59.3        | 78.1        | 79.8        | 65.4        | 75.5        | 76.4        | 65.7        | 58.0        | 81.0        | 72.0        | 64.4        | 83.3        | 71.6               |
| U-SFAN [35]              | 57.8        | 77.8        | 81.6        | 67.9        | 77.3        | 79.2        | 67.2        | 54.7        | 81.2        | 73.3        | 60.3        | 83.9        | 71.9               |
| CoWA-JMDS [21]           | 56.9        | 78.4        | 81.0        | 69.1        | <u>80.0</u> | <b>79.9</b> | <u>67.7</u> | 57.2        | 82.4        | 72.8        | 60.5        | 84.5        | 72.5               |
| DaC [57]                 | 59.1        | 79.5        | 81.2        | <u>69.3</u> | 78.9        | 79.2        | 67.4        | 56.4        | 82.4        | 74.0        | <b>61.4</b> | 84.4        | 72.8               |
| C-SFDA [19]              | <u>60.3</u> | <u>80.2</u> | <b>82.9</b> | <u>69.3</u> | <b>80.1</b> | 78.8        | 67.3        | <u>58.1</u> | <b>83.4</b> | 73.6        | <u>61.3</u> | <u>86.3</u> | <b>73.5</b>        |
| AaD [53]                 | 59.3        | 79.3        | <u>82.1</u> | 68.9        | 79.8        | 79.5        | 67.2        | 57.4        | <u>83.1</u> | 72.1        | 58.5        | 85.4        | 72.7               |
| <b>Improved SFDA</b>     | <b>60.7</b> | 78.9        | 82.0        | <b>69.9</b> | 79.5        | <u>79.7</u> | 67.1        | <b>58.8</b> | 82.3        | <b>74.2</b> | <u>61.3</u> | <b>86.4</b> | <u>73.4 (+0.7)</u> |

Table 4. **Classwise Accuracy (%) on VisDA2017 (ResNet-101)**. The best and second best are highlighted in **bold** and with underline.

| Method (Synthetic → Real) | plane       | bicycl      | bus         | car         | horse       | knife       | mcycl       | person      | plant       | sktbrd      | train       | truck       | Per-class          |
|---------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------------|
| 3C-GAN [22]               | 94.8        | 73.4        | 68.8        | 74.8        | 93.1        | 95.4        | 88.6        | 84.7        | 89.1        | 84.7        | 83.5        | 48.1        | 81.6               |
| SHOT [23]                 | 94.3        | 88.5        | 80.1        | 57.3        | 93.1        | 94.9        | 80.7        | 80.3        | 91.5        | 89.1        | 86.3        | 58.2        | 82.9               |
| VDM-DA [42]               | 96.9        | 89.1        | 79.1        | 66.5        | 95.7        | 96.8        | 85.4        | 83.3        | 96.0        | 86.6        | 89.5        | 56.3        | 85.1               |
| A <sup>2</sup> Net [49]   | 94.0        | 87.8        | 85.6        | 66.8        | 93.7        | 95.1        | 85.8        | 81.2        | 91.6        | 88.2        | 86.5        | 56.0        | 84.3               |
| G-SFDA [52]               | 96.1        | 83.3        | 85.5        | 74.1        | 97.1        | 95.4        | 89.5        | 79.4        | 95.4        | 92.9        | 89.1        | 42.6        | 85.4               |
| NRC [51]                  | 96.8        | <u>91.3</u> | 82.4        | 62.4        | 96.2        | 95.9        | 86.1        | 80.6        | 94.8        | 94.1        | 90.4        | 59.7        | 85.9               |
| CPGA [33]                 | 95.6        | 89.0        | 75.4        | 64.9        | 91.7        | <b>97.5</b> | 89.7        | 83.8        | 93.9        | 93.4        | 87.7        | <b>69.0</b> | 86.0               |
| U-SFAN [35]               | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | 82.7               |
| AdaContrast [5]           | 97.0        | 84.7        | 84.0        | 77.3        | 96.7        | 93.8        | 91.9        | <u>84.8</u> | 94.3        | 93.1        | <b>94.1</b> | 47.9        | 86.8               |
| CoWA-JMDS [21]            | 96.2        | 89.7        | 83.9        | 73.8        | 96.4        | <u>97.4</u> | 89.3        | <b>86.8</b> | 94.6        | 92.1        | 88.7        | 53.8        | 86.9               |
| DaC [57]                  | 96.6        | 86.8        | <u>86.4</u> | 78.4        | 96.4        | 96.2        | <b>93.6</b> | 83.8        | <b>96.8</b> | <b>95.1</b> | 89.6        | 50.0        | 87.3               |
| C-SFDA [19]               | <b>97.6</b> | 88.8        | 86.1        | 72.2        | <u>97.2</u> | 94.4        | 92.1        | 84.7        | 93.0        | 90.7        | <u>93.1</u> | 63.5        | 87.8               |
| AaD [53]                  | 97.4        | 90.5        | 80.8        | 76.2        | <b>97.3</b> | 96.1        | 89.8        | 82.9        | 95.5        | 93.0        | 92.0        | <u>64.7</u> | <u>88.0</u>        |
| <b>Improved SFDA</b>      | <u>97.5</u> | <b>91.4</b> | <b>87.9</b> | <b>79.4</b> | <u>97.2</u> | 97.2        | <u>92.2</u> | 83.0        | <u>96.4</u> | <u>94.2</u> | 91.1        | 53.0        | <b>88.4 (+0.4)</b> |

Table 5. **Analysis of Augmentation Training.**

| Init. Aug       | Aug Training | Accuracy [%]        | $\Delta$ AaD          |
|-----------------|--------------|---------------------|-----------------------|
| Random          | ✓            | 73.1<br><b>73.3</b> | + 0.4<br><b>+ 0.6</b> |
| AutoAugment [7] | ✓            | 73.3<br><b>73.4</b> | + 0.6<br><b>+ 0.7</b> |

Table 6. **Analysis of Auto-Adjusting Diversity Constraint.**

|                                     | $\lambda_{\text{div}}^{\text{max}}$ | Accuracy [%] |
|-------------------------------------|-------------------------------------|--------------|
| Fixed Diversity Constraint          | 0.1                                 | 69.3         |
|                                     | 0.25                                | 72.8         |
|                                     | 0.5                                 | 72.6         |
|                                     | 0.75                                | 71.1         |
| Auto-Adjusting Diversity Constraint | 0.75                                | <b>73.4</b>  |

## 5.2. Main results

We evaluated the performance of our method by taking the average score of three different runs for all benchmarks.

**Result on Office-31.** The results are shown in Tab. 2. Our method improved accuracy by 0.4% on average compared with the baseline AaD [53]. Furthermore, ours was comparable in accuracy to the second best method, CoWA-JMDS [21] and only 0.2% off the best method C-SFDA [19].

**Result on Office-Home.** The results are shown in Tab. 3. Ours improved accuracy by 0.7% on average compared with AaD [53] and was second best. The accuracy difference from the best method, C-SFDA [19], was merely 0.1%.

**Result on VisDA2017.** The results are shown in Tab. 4. Ours improved accuracy by 0.4% compared with AaD [53]. The average of per-class accuracy reached 88.4%, which was the best among the compared methods.

## 5.3. Analysis

**Ablation study.** Using *Office-Home*, we analyzed the effectiveness of the proposed two components.

**Augmentation Training with Teacher-Student Framework** was validated by comparing its performance with that of a fixed augmentation variant. The results are shown in Tab. 5. When starting with the randomly initialized data augmentation, our augmentation training yielded a larger accuracy gain from the base method ( $\Delta$ AaD), 1.5 times greater than without augmentation training. The accuracy gain is slightly smaller when the augmentation is initialized with AutoAugment policies, but ours still yielded better results. More detailed results are provided in Appendix C.1. **Model Training with Auto-Adjusting Diversity Constraint** was validated by comparing our method with a variant that uses a fixed  $\lambda_{\text{div}}$ , i.e.,  $\lambda_{\text{div}} = \lambda_{\text{div}}^{\text{max}}$ . Considering that the optimal  $\lambda_{\text{div}}^{\text{max}}$  for this “Fixed Diversity Constraint” may differ from that of our method, we performed experiments with several values of  $\lambda_{\text{div}}^{\text{max}}$ . The results are shown in

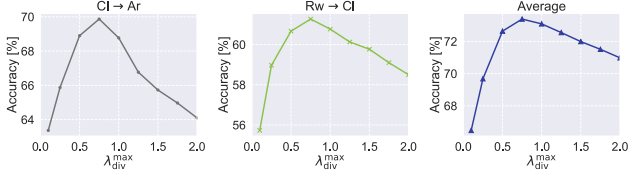


Figure 4. Analysis of the coefficient parameter  $\lambda_{\text{div}}^{\text{max}}$

Table 7. Analysis of Application to SHOT-IM.

|               | Office-31          | Office-Home        | VisDA2017          |
|---------------|--------------------|--------------------|--------------------|
| SHOT-IM [23]  | 87.3               | 70.5               | 80.4               |
| Improved SFDA | <b>88.6 (+1.3)</b> | <b>71.2 (+0.7)</b> | <b>83.3 (+2.9)</b> |

Tab. 6. Our method outperformed all fixed  $\lambda_{\text{div}}$  variants, which indicates the validity of our proposed techniques. We also obtained the same results from the synthetic data experiment, which is described in Appendix B.2.

**Applicability to other SFDA methods.** Although we based on the implementation of our method on AaD, the theoretical insights and techniques we have made here should be applicable to many other SFDA methods. To confirm this, we empirically verified the applicability of our proposed techniques to another SFDA method, SHOT-IM [23]. Here, we used the same hyper-parameters as in Sec. 5.1, except for the following points; we set  $\lambda_{\text{div}}^{\text{max}}$  to 0.7 and  $\eta$  to  $2e-3$  for *Office-31* and *Office-Home*,  $\eta_{\text{init}}$  to  $1e-3$  and  $\lambda_{\text{div}}^{\text{max}}$  to 0.8 for *VisDA2017*.

The results are shown in Tab. 7. We can see a steady improvement in all the benchmarks, which demonstrate the effectiveness of our techniques for SHOT-IM. Moreover, since many of the existing SFDA methods are built upon SHOT-IM, this result implies that our method is valid for a wider range of SFDA methods.

**Hyper-parameter analysis.** We conducted experiments using *Office-Home* to analyze the effect of the parameter  $\lambda_{\text{div}}^{\text{max}}$  in Model Training with Auto-Adjusting Diversity Constraint, and on the parameters  $\hat{e}$  and  $\lambda_A$  in Augmentation Training with Teacher-Student Framework.

**Analysis of  $\lambda_{\text{div}}^{\text{max}}$  (Fig. 4)**  $\lambda_{\text{div}}^{\text{max}}$  controls the strength of the prediction diversity constraint. We varied  $\lambda_{\text{div}}^{\text{max}}$  from 0.0 to 2.0 and evaluated the accuracy.  $\lambda_{\text{div}}^{\text{max}}$  is optimal at 0.75, and the accuracy deteriorates if it is larger or smaller than the optimal value. In particular, the accuracy deteriorates more sharply when it takes a smaller value than a larger value. This result is in line with our theoretical analysis. Specifically, if  $\lambda_{\text{div}}^{\text{max}}$  is too small, the diversity loss will not play the role of constraining the objective (3) sufficiently, and thus we can not obtain the upper bound for the target error. However, if it is too large, the diversity loss becomes an excessive constraint, and that reduces the accuracy.

**Analysis of  $\hat{e}$  (Fig. 5).**  $\hat{e}$  controls the frequency of the augmentation training. We analyzed the effect of  $\hat{e}$  by varying it from one to six. The optimal value of  $\hat{e}$  is two or three and

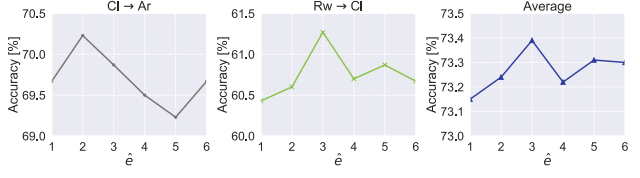


Figure 5. Analysis of the interval parameter  $\hat{e}$

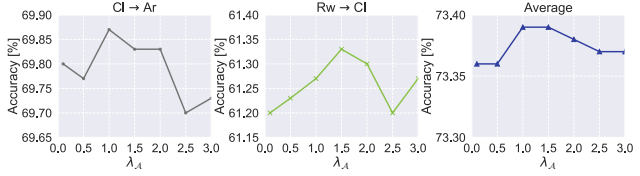


Figure 6. Analysis of the coefficient parameter  $\lambda_A$

the performance is lower if it is set to a larger or smaller value than this. Our augmentation training will not demonstrate its validity unless the current model and the teacher model are different to some extent. When  $\hat{e}$  is small, the teacher and the current model are too close to exhibit the full potential of the augmentation training, while when  $\hat{e}$  is large, the data augmentation is not trained well enough.

**Analysis of  $\lambda_A$  (Fig. 6).**  $\lambda_A$  controls the effect of  $\mathcal{L}_{\text{Student}}$  and  $\mathcal{L}_{\text{Teacher}}$ , where  $\mathcal{L}_{\text{Student}}$  increases  $d$  of the bound (4) by encouraging the augmentation to generate more difficult samples while  $\mathcal{L}_{\text{Teacher}}$  decreases  $\mu$  of the bound (4) by encouraging the augmentation to keep the semantics of the data. We analyzed how our method performed while varying  $\lambda_A$  from 0.0 to 3.0. The optimal value of  $\lambda_A$  is 1.0 to 1.5. If the balance is not proper, it will cause a decrease in  $d$  or an increase in  $\mu$ , resulting in the prediction model having poor accuracy. The results thus demonstrate that our method is consistent with our theoretical insights.

## 6. Conclusion

We shed light on the theoretical perspective of existing SFDA methods through the theory of self-training based on the expansion assumption [48]. Our finding that the SFDA training loss with discriminability and diversity functions the same way as the training objective of the theory not only provided a way to understand existing SFDA methods but also yielded two novel techniques for improving the performance of SFDA methods. The experimental results and in-depth analysis justified the validity of our theoretical insights and proposed method. We expect that this work will encourage further development of SFDA research.

On the other hand, since this study aims to understand existing SFDA methods, we leave one limitation. That is, we cannot take into account how well the source model originally performs on the target domain, which is also not considered in existing SFDA research. One of our future works is to develop a method that overcomes this weakness and advances SFDA to be more practical.



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